

Phase 1: System Identification Report

System Selected: Automated EOT Crane

Category: Category D - Mechanical & Industrial Systems

1. Real-World Application

EOT cranes are heavily utilized in industrial warehouses and manufacturing floors to safely lift and transport heavy payloads across three dimensions. Understanding the dynamics is critical for implementing anti-sway control algorithms that prevent dangerous payload oscillations during sudden stops or starts.

2. System Variables & Parameters

1. INPUTS:

- Input 1: F_x (Force applied by the bridge motor)
- Input 2: F_y (Force applied by the trolley motor)
- Input 3: F_z (Hoisting tension force altering the cable length)

2. OUTPUTS:

- Output 1: Position of the payload in 3D space (x_p, y_p, z_p)
- Output 2: Swing angles of the payload

3. STATE VARIABLES:

- State Variable 1: x_b (Position of the bridge)
- State Variable 2: y_t (Position of the trolley on the bridge)
- State Variable 3: L (Length of the hoist cable)
- State Variable 4: θ_x and θ_y (Pendulum swing angles of the payload)
- State Variable 5: Velocities for the bridge, trolley, cable length, and swing angles

4. SYSTEM PARAMETERS:

- System Parameter 1: M_b (Mass of the bridge)
- System Parameter 2: m_t (Mass of the trolley)
- System Parameter 3: m_p (Mass of the payload)
- System Parameter 4: g (Acceleration due to gravity)

3. Degrees of Freedom (DOF) Identification

The full spatial system possesses 5 Degrees of Freedom.

1. Bridge translation along the X-axis.
2. Trolley translation along the Y-axis.
3. Hoist cable extension along the Z-axis.
4. Payload swing angle θ_x (rotation about the Y-axis).
5. Payload swing angle θ_y (rotation about the X-axis).

4. Key Assumptions & Physical Reasoning

- Rigid Body Dynamics: The bridge and trolley are modelled as rigid bodies with no flex or deformation.
- Particle Kinetics for Payload: The payload is modelled as a point mass particle.
- Massless & Inextensible Cable: The hoist cable does not stretch, and its mass is negligible compared to the payload.
- Viscous Friction: Friction between the wheels and rails is proportional to velocity.

5. Coordinate System & Sign Convention

A fixed Cartesian coordinate system (X, Y, Z) is established at the corner of the runway.

- Positive X: Forward along the runway.
- Positive Y: Rightward across the bridge span.
- Positive Z: Downward (direction of gravity).
- Swing angles are measured from the vertical Z-axis, with counter-clockwise taken as positive.

6. Free Body Diagram (FBD) Description



